

CHAPTER 1

Units and Measurements

Systems of Units

- The angle of 1' (minute of arc) in radian is nearly equal to, (2020-Covid)
 - 4.85×10^{-4} rad
 - 4.80×10^{-6} rad
 - 1.75×10^{-2} rad
 - 2.91×10^{-4} rad
- The unit of thermal conductivity is (2019)
 - $\text{W m}^{-1} \text{K}^{-1}$
 - J m K^{-1}
 - $\text{J m}^{-1} \text{K}^{-1}$
 - W m K^{-1}

Dimensions of Physical Quantities

- The quantities which have the same dimensions as those of solid angle are: (2024)
 - strain and arc
 - angular speed and stress
 - strain and angle
 - stress and angle
- A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at given time t . The factor which is dimensionless, if α and β are constants, is: (2024)
 - $\alpha\beta t$
 - $\alpha\beta/t$
 - $\beta t/\alpha$
 - α/β
- Plane angle and solid angle have: (2022)
 - Both units and dimension
 - Units but no dimensions
 - Dimensions but no units
 - No units and no dimensions
- The dimension $[\text{MLT}^{-2}\text{A}^{-2}]$ belong to the: (2022)
 - electric permittivity
 - magnetic flux
 - self inductance
 - magnetic permeability
- If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of: (2021)
 - $[\text{M}][\text{L}^{-1}][\text{T}^{-1}]$
 - $[\text{M}][\text{L}^0][\text{T}^0]$
 - $[\text{M}^2][\text{L}^{-2}][\text{T}^{-1}]$
 - $[\text{M}^2][\text{L}^{-1}][\text{T}^0]$
- Dimensions of stress are: (2020)
 - $[\text{ML}^2\text{T}^{-2}]$
 - $[\text{ML}^0\text{T}^{-2}]$
 - $[\text{ML}^{-1}\text{T}^{-2}]$
 - $[\text{MLT}^{-2}]$

Application of Dimensions

- If force $[\text{F}]$, acceleration $[\text{A}]$ and time $[\text{T}]$ are chosen as the fundamental physical quantities. Find the dimensions of energy. (2021)
 - $[\text{F}][\text{A}][\text{T}^2]$
 - $[\text{F}][\text{A}][\text{T}^{-1}]$
 - $[\text{F}][\text{A}^{-1}][\text{T}]$
 - $[\text{F}][\text{A}][\text{T}]$

- A physical quantity of the dimensions of length that can be formed out of c , G and $\frac{e^2}{4\pi\epsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]: (2017-Delhi)

- $c^2 \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$
- $\frac{1}{c^2} \left[\frac{e^2}{G 4\pi\epsilon_0} \right]^{1/2}$
- $\frac{1}{c^2} G \frac{e^2}{4\pi\epsilon_0}$
- $\frac{1}{c^2} \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$

- Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length? (2016-II)

- $\sqrt{\frac{hc}{G}}$
- $\sqrt{\frac{Gc}{h^{3/2}}}$
- $\frac{\sqrt{hG}}{c^{3/2}}$
- $\sqrt{\frac{hG}{c^{5/2}}}$

- If energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be: (2015)
 - $[\text{EV}^{-1}\text{T}^{-2}]$
 - $[\text{EV}^{-2}\text{T}^{-2}]$
 - $[\text{E}^{-2}\text{V}^{-1}\text{T}^{-3}]$
 - $[\text{EV}^{-2}\text{T}^{-1}]$
- If dimension of critical velocity of liquid flowing through a tube are expressed as $v_c \propto [\eta^x \rho^y r^z]$ where η , ρ and r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x , y and z are given by: (2015 - Re)
 - 1, 1, 1
 - 1, -1, -1
 - 1, -1, 1
 - 1, -1, -1
- If Force (F), Velocity (V) and Time (T) are taken as fundamental units, then the dimensions of mass are: (2014)
 - $[\text{F V T}^{-1}]$
 - $[\text{F V T}^{-2}]$
 - $[\text{F V}^{-1} \text{T}^{-1}]$
 - $[\text{F V}^{-1} \text{T}]$

Errors

- The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are: (2023)
 - Random errors
 - Instrumental errors
 - Personal errors
 - Least count errors
- A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be: (2023)
 - 1.4%
 - 1.2%
 - 1.3%
 - 1.6%

17. The percentage error in the measurement of g is:

(Given that $g = \frac{4\pi^2 L}{T^2}$, $L (10 \pm 0.1) \text{ cm}$, $T = (100 \pm 1) \text{ s}$)
(2022 Re)

- a. 7% b. 2% c. 5% d. 3%
18. The intervals measured by a clock given the following readings: 1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s. What is the percentage relative error in the observations? (2020-Covid)
- a. 4% b. 16% c. 1.6% d. 2%
19. In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement of X, where $X = \frac{A^2 B^{1/2}}{C^{1/3} D^3}$ will be (2019)
- a. $\left(\frac{3}{13}\right)\%$ b. 16% c. -10% d. 10%
20. In an experiment four quantities a, b, c and d are measured with percentage error 1%, 2%, 3% and 4% respectively. Quantity P is calculated as follows $P = \frac{a^3 b^2}{cd}$. % error in P is: (2013)
- a. 4% b. 14% c. 10% d. 7%

Significant Figures

21. The diameter of a spherical bob, when measured with vernier callipers yielded the following values: 3.33 cm, 3.32 cm, 3.34 cm, 3.33 cm and 3.32 cm. The mean diameter to appropriate significant figures is: (2023-Manipur)
- a. 3.328 cm b. 3.3 cm c. 3.33 cm d. 3.32 cm
22. The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is: (2022)
- a. 14×10^2 b. 138×10^1
c. 1382 d. 1382.5

23. Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$? (2020)
- a. 9.98 m b. 9.980 m c. 9.9 m d. 9.9801 m

Measuring Instruments

24. The pitch of an error free screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a thick wire, the pitch scale reads 1 mm and 63rd division on the circular scale coincides with the reference line. The diameter of the wire is: (2024 Re)
- a. 1.63 cm b. 0.163 cm c. 0.163 m d. 1.63 m
25. In a vernier calipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is: (2024)
- a. 100 N b. $10(N + 1)$
c. $\frac{1}{10N}$ d. $\frac{1}{100(N + 1)}$
26. A screw gauge gives the following readings when used to measure the diameter of a wire
Main scale reading: 0 mm
Circular scale reading: 52 divisions
Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is: (2021)
- a. 0.026 cm b. 0.26 cm c. 0.052 cm d. 0.52 cm
27. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale.
The pitch of the screw gauge is: (2020)
- a. 0.25 mm b. 0.5 mm c. 1.0 mm d. 0.01 mm
28. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of -0.004 cm, the correct diameter of the ball is (2018)
- a. 0.053 cm b. 0.525 cm c. 0.521 cm d. 0.529 cm

Answer Key

1. (d) 2. (a) 3. (c) 4. (d) 5. (b) 6. (d) 7. (d) 8. (c) 9. (a) 10. (d)
11. (c) 12. (b) 13. (b) 14. (d) 15. (a) 16. (d) 17. (d) 18. (c) 19. (b) 20. (b)
21. (c) 22. (a) 23. (a) 24. (b) 25. (d) 26. (c) 27. (b) 28. (d)